ICE 5.1

Aman Gupta • Feb 11, 2025 (Edited Feb 21, 2025)

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1. (2 marks) Assume we have an 8-bit linear feedback shift register being used to generate a PRNG stream. What is the maximum periodicity one can expect?

- 8
- **255 (correct)**
- 256
- None of the above

2. (3 marks) Assume we are using the linear congruential PRNG to generate a byte stream: X(n+1) = a * X(n) mod m, with m = 19, a = 4, and X(0) = 3. What is the periodicity on this PRNG generator? Hint go ahead and compute the sequence of random numbers generated, possibly using the initial X(0) = 3.

- 4
- 9
- **18 (correct)**
- 19
- None of the above

3. (3 marks) Take a look at the 16 bit shift-register based PRNG below. If at a point the contents of the register is 0xACE0 (as given below), then what is the next random number generated by the PRNG?

- 0x5671
- 0xACE0
- **0xD670 (correct)**
- None of the above

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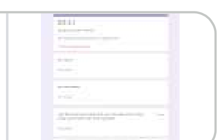

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